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Paleomagnetic Study of the Cretaceous Atajaña Formation and the Arica Dike Swarm, Northernmost Chile

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Remanent magnetization directions were obtained from 28 red sandstone samples of the Lower Cretaceous Atajaña Formation. Paleomagnetic studies were also performed on 103 igneous rock samples taken from 19 dikes in the Arica dike swarm, which intrudes into the Atajaña Formation sandstone. They showed 10°–15° counterclockwise rotation of the Arica region, northernmost Chile, with respect to the stable South American platform. These data suggest that the northernmost Chilean region underwent a tectonic rotation at least after the Cretaceous period, as the southern end of the Peruvian block.

1. Introduction

The recent paleomagnetic studies of the Central Andean area (PALMER *et al.*, 1980; HEKI, 1983; HEKI *et al.*, 1983, 1984, 1985a, b; KONO *et al.*, 1984) are revealing that the Peruvian Andes has rotated counterclockwise as a nearly rigid "block" giving rise to the present elbow shape of the Andes at the Peru-Chile border (Bolivian orocline; CAREY, 1955). The northernmost part of Chile (the Arica region) is an area of paleomagnetic importance because it accurately corresponds to the "hinge" of the Bolivian orocline. In this article, we discuss the Cretaceous paleomagnetic data of this region obtained from a red sandstone layer of the Atajaña Formation and from a dike swarm (the Arica dike swarm) intruding into this formation.

2. Geology

The geological information of the Arica Department (Departamento de Arica) is available in SALAS *et al.* (1966). The Jurassic units in the region (called the Arica Group) are overlain unconformably by the Neocomian Vilacollo Group.

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Units of this group exposed in the coastal region (the Coastal Cordillera) are called the Atajaña Formation, which is composed of andesitic volcanics and terrestrial clastic sediments and is overlain unconformably by the Tertiary Azapa Formation or the Oxaya Formation. The Atajaña Formation was first named by CECIONI and GARCIA (1960) for the rocks of the Atajaña Mountain in the Departamento de Pisagua, south of the Departamento de Arica. In the coastal region of the Departamento de Arica, lithologically very similar rock sequences overlying the Arica Group were found and were suggested to correspond to the Atajaña Formation by SALAS *et al.* (1966).

The outcrops of the sedimentary sequences of this formation along Quebrada Vitor (Vitor Canyon), some 30 km south of Arica City, correspond to the middle member of the Atajaña Formation (SALAS *et al.*, 1966). Twenty-eight oriented hand samples were taken for a paleomagnetic study from several different horizons of the well stratified coarse reddish violet sandstone layer in this exposure (Fig. 1, AR24).

Along Quebrada de la Higuera, some 10km southeast of the city of Arica, we found basaltic and andesitic dikes intruding into the red sandstone and tuff breccia layers of the Atajaña Formation. These dikes were named "the Arica dike swarm" by us (Fig. 1). The dikes generally trend east-west, which is almost parallel with the present direction of the plate convergence at the Peru-Chile trench. The typical thicknesses of the dikes are 1–2m. Five to seven oriented hand samples were taken from each of 19 dikes (AR31–49) which were exposed along the Panamerican Highway. At present, radiometric age data of the dikes are absent and we are not 100% sure that the dikes are of Cretaceous age. However, this uncertainty does not affect the discussion on the amount of the tectonic movements of this region after the intrustion time because the paleomagnetic pole of the whole of South America is almost stationary during the Mesozoic and Cenozoic times (HEKI, 1983). Hence we tentatively assumed the Cretaceous age of the dikes in the subsequent discussions.

We also took paleomagnetic samples from the sandstone layer of the country rock (AR50) in order to compare them with AR24. The AR50 samples are sufficiently far from the adjacent dikes and no thermal effects of the dikes are considered to be present for these samples.

3. Experimental Procedure and Paleomagnetic Results

3.1 Red sandstone (AR24)

Magnetic remanences of AR24 red sandstones were measured using a Schonstedt spinner magnetometer at the University of Tokyo. Original natural remanent magnetization (NRM) intensities are mostly around 1×10^{-5} Am^2/kg. At first, stepwise alternating field (AF) demagnetization was carried out but little change of the direction/intensity of the remanent magnetization was observed. Therefore, stepwise thermal demagnetization was performed on each specimen. The blocking temperature distribution was found to be ranging up to over 650°C

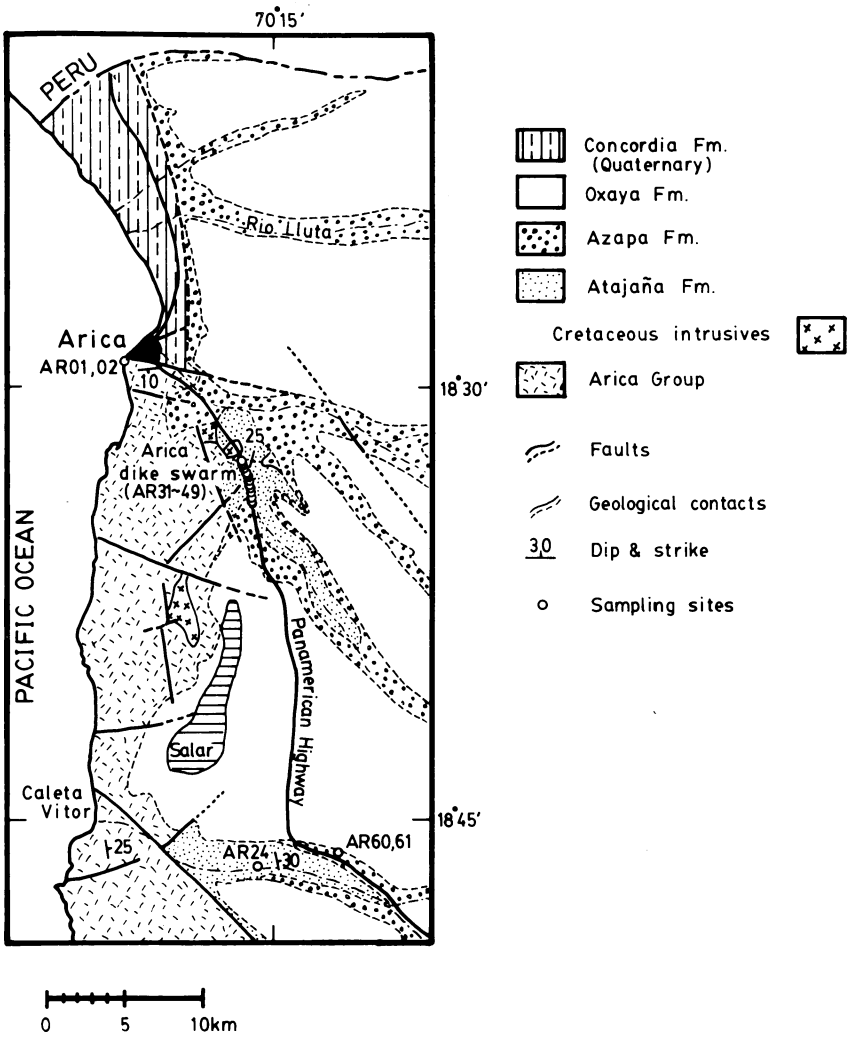


Fig. 1. Geologic map (SALAS *et al.*, 1966) and the sampling localities for the red sandstone of the Atajaña Formation (AR24) and for the Arica dike swarm (AR31–49). The latitude and the longitude on the map are northern and western, respectively.

suggesting the hematite as the major carrier of the remanence. The Zijderveld demagnetization diagram (ZIJDERVELD, 1967) indicates that the NRM consists of the stable single-component remanent magnetization (Fig. 2). The remanent magnetization directions at the 600°C step were taken as the paleofield directions and then were corrected for the bedding plane (N10°W, 25°E). All specimens showed normal polarity directions and we found no significant direction/intensi-

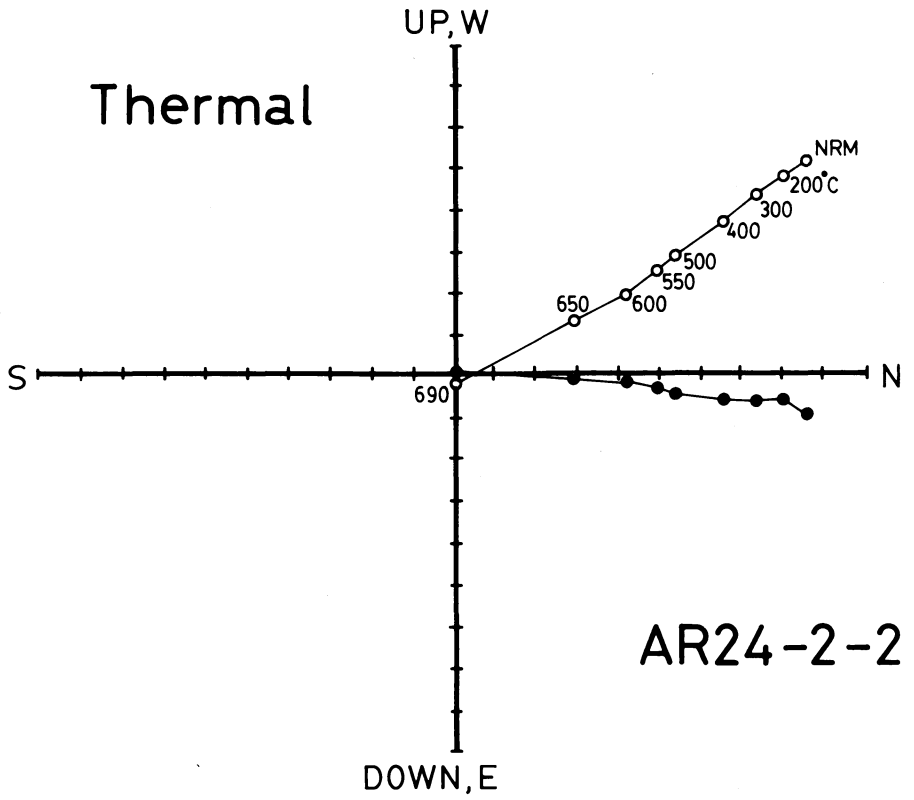


Fig. 2. Zijderveld diagram in the progressive thermal demagnetization of the red sandstone specimen of the Atajaña Formation (AR24-2-2). Open and solid symbols denote projections onto vertical and horizontal planes respectively. Demagnetizing temperatures (°C) are also shown. Scales are arbitrarily taken.

ty differences among the sampled horizons. They showed slight counterclockwise declination shift from the present north. Paleomagnetic results are listed in Table 1 and illustrated in Fig. 3.

3.2 Arica dike swarm (AR31-49) and the country rock (AR50)

A spinner magnetometer was also used for the measurement of these rocks. The dike specimens presented the NRM intensities of the order of 10^{-5} Am^2/kg. Stepwise AF demagnetization was performed on each specimen. Median destructive fields (MDFs) were between 10mT and 15mT for most specimens. The Zijderveld diagram (Fig. 4) demonstrates that the NRM consists of a stable single-component magnetization with a minor amount of secondary overprints. Thermal demagnetization was also performed on several specimens and also single-component magnetization, with the blocking temperatures ranging from 350°C

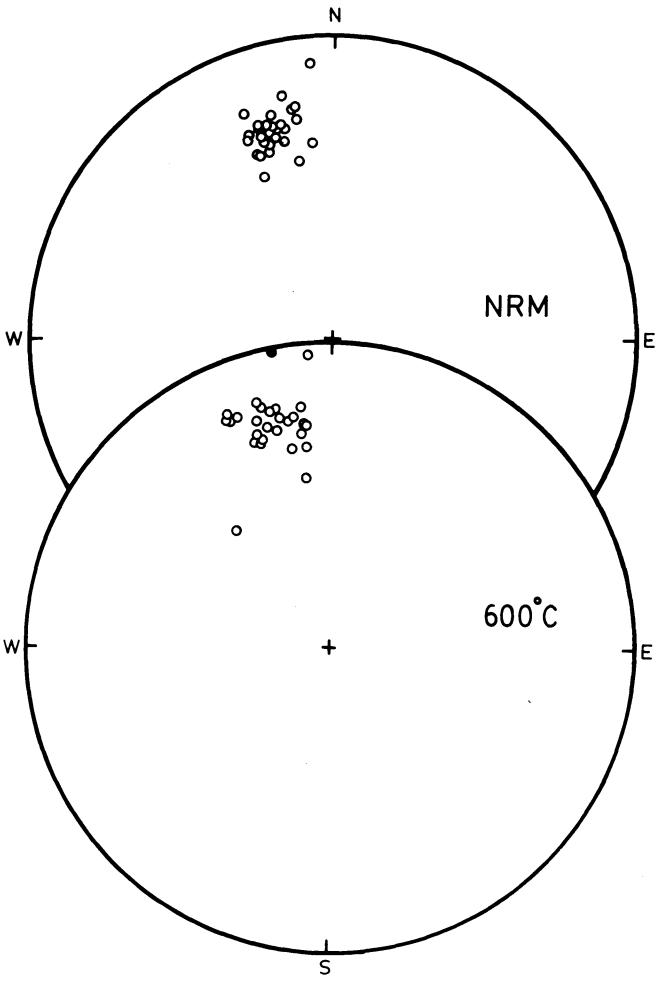


Fig. 3. Lambert equal-area projection of the remanent magnetization directions of AR24 at the NRM step and the 600°C step. Open and solid circles indicate negative (upward) and positive (downward) inclinations respectively.

to 580°C, was identified. Paleomagnetic directions were determined from the gradients of the linear portions of such demagnetization diagrams by the least squares fitting. For two dikes (AR46, 49), however, the optimum demagnetization steps were determined by the minimum dispersion criterion because they did not present a sufficient length of the linear portion due to relatively large secondary overprints.

The *in situ* dike contacts are almost vertical and the bedding correction using the bedding plane of the host rock (N60°E, 15°E) makes these contacts plunge

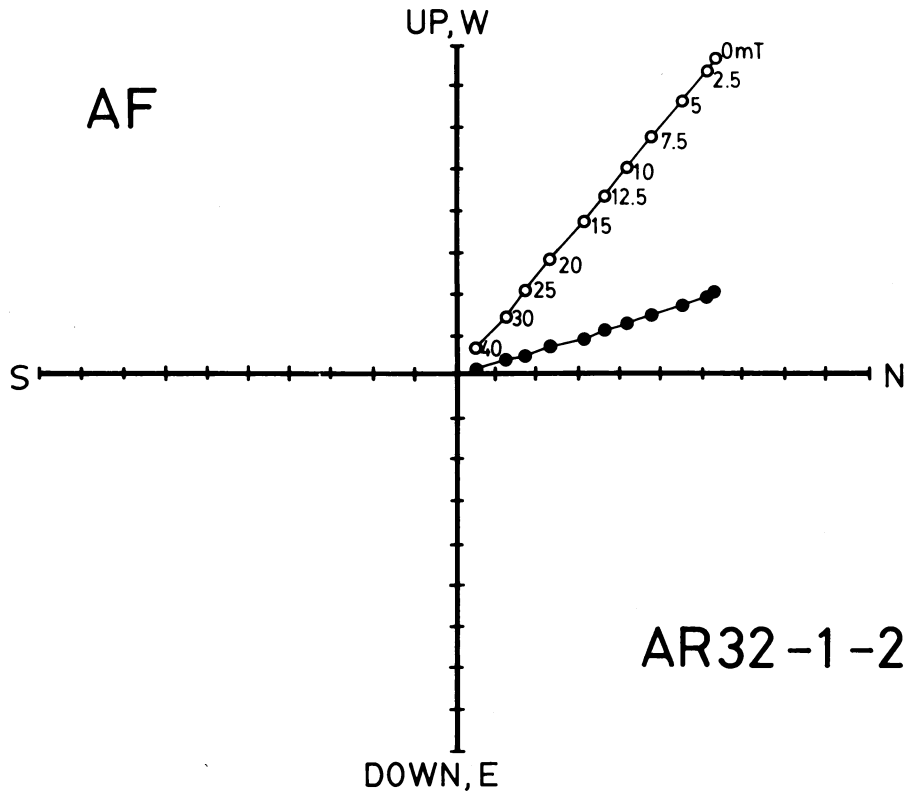


Fig. 4. Zijderveld diagram of the Arica dike swarm specimen (AR32-1-2) in the progressive AF demagnetization. Demagnetizing AF intensities (mT, peak intensity) are also shown.

systematically toward a certain direction. Hence, the dikes are thought to have intruded after the tilting of the country rock and no structural corrections were made for the remanent magnetization directions. The obtained remanent magnetization directions showed considerably good within-dike consistency. All the directions were of normal polarity and slightly deviate counterclockwise from the present axial dipole field, which is consistent with the result of AR24 sandstone. The dike-mean directions and the 95% confidence circles (FISHER, 1953) are illustrated in Fig. 5 and listed in Table 1.

For the host rock samples (AR50), both AF and thermal demagnetizations were carried out and the stable remanent magnetization components were isolated. The obtained average direction after the structural correction (the bedding plane: N60°E, 15°E) shows good coincidence with that of AR24 red sandstone (Table 1). This suggests that the acquisition of the stable remanences of AR24 and AR50 are pre-folding.

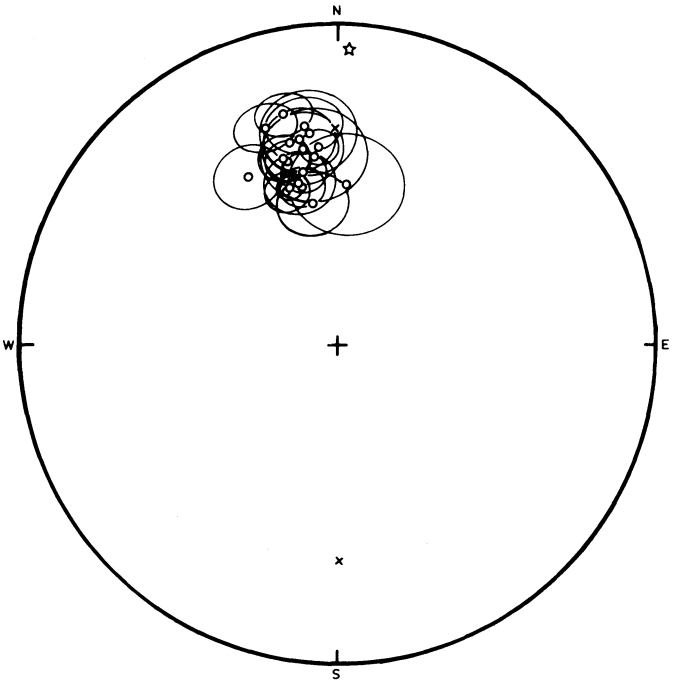


Fig. 5. Lambert equal-area projection of the dike-mean remanent magnetization directions of the Arica dike swarm (AR31–49). Ninty-five percent confidence circles after FISHER (1953) are also illustrated. The star indicates the present field direction in the Arica region. The Xs show the present axial dipole field directions.

4. Discussion

The remanent magnetization directions of 28 specimens of AR24 were averaged and the pole corresponding to this mean direction was calculated. The virtual geomagnetic poles (VGPs) corresponding to the mean directions of the 19 dikes of the Arica dike swarm were also averaged to obtain the paleomagnetic pole. Table 2 summarizes these Cretaceous paleomagnetic poles obtained in the Arica region.

As the reference Cretaceous paleomagnetic poles of the stable South American platform, we have selected four paleomagnetic poles with objective criteria (see HEKI *et al.*, 1984) from the following paleomagnetic data: the Serra Geral Formation (CREER, 1962; PACCA and HIODO, 1976), Poços de Caldas Complex (OPDYKE and MACDONALD, 1973), Maranhâo basalt intrusions (SCHULT and GUERREIRO, 1979) and the Cabo de Sto. Agostinho volcanics (SCHULT and GUERREIRO, 1980). Figure 6 shows the Cretaceous paleomagnetic poles of the Arica region (circles) and the stable platform (stars) as the southern hemisphere poles.

Table 1. Paleomagnetic directional data of the Cretaceous rocks in the Arica region.

Site	*N*	Incl.* (°)	Decl.* (°)	R	*k*	α_{95} (°)	Pole Lat. (°N)	Pole Long. (°E)
			(the Atajaña Formation, red sandstone)					
AR24**	28	−25.6 (−25.1,	−14.9 −2.8)	27.4089	46	4.1	74.7	−142.1
			(the Arica dike swarm)					
AR31	5	−39.0	−5.3	4.9825	228	5.1	84.0	164.1
AR32	7	−45.0	−11.6	6.9526	127	5.4	76.7	160.9
AR33	5	−36.1	−13.8	4.9709	138	6.5	76.9	−168.8
AR34	5	−32.6	−8.8	4.9508	81	8.5	81.6	−155.8
AR35	6	−41.0	−15.4	5.9646	141	5.7	74.8	178.2
AR36	3	−48.1	−17.0	2.9986	1388	3.3	71.3	162.4
AR37	5	−38.7	−10.0	4.9448	73	10.0	80.1	179.0
AR38	6	−41.0	−28.1	5.9161	60	8.8	63.4	−175.8
AR39	5	−30.4	−18.8	4.9646	113	7.2	71.9	−156.1
AR40	6	−53.5	−10.2	5.9216	64	8.5	72.1	138.2
AR41	6	−40.0	−16.1	5.9567	116	6.3	74.4	−178.5
AR42	4	−41.4	−7.1	3.9355	47	13.6	81.6	160.4
AR43	6	−47.9	−13.6	5.9285	70	8.1	73.8	157.3
AR44	6	−48.9	−13.2	5.9285	146	5.6	73.6	154.2
AR45	5	−49.4	3.1	4.8797	33	13.5	78.0	96.7
AR46	6	−36.1	−10.5	5.9054	53	9.3	80.0	−170.2
AR47	6	−48.8	−18.2	5.9690	161	5.3	70.1	162.4
AR48	6	−28.0	−13.4	5.9532	107	6.5	76.6	−146.2
AR49	5	−34.9	−7.7	4.9180	49	11.1	82.7	−166.5
			(the country rock)					
AR50**	6	−27.9 (−42.8,	−23.3 −21.9)	5.9873	395	3.4	67.4	−154.2

N: number of samples studied, R: length of the resultant vector, *k*: precision parameter, α_{95}: radius of 95% confidence circle.

*Remanent magnetization directions were determined by the least squares fitting to the linear portion of the demagnetization diagram except for AR24 (600°C), AR46 (10 mT), AR49 (12.5 mT) and AR50 (200 mT). See text.

**Remanent magnetization directions were structurally corrected. (the *in situ* directions are shown in the brackets).

These two pole groups are significantly discordant, which suggests that the Arica region underwent post-Cretaceous tectonic movement with respect to the stable platform. Such discordances of the poles in the orogenic zones are usually described in terms of "rotation" and "flattening" of the paleomagnetic directions (BECK, 1980). These values represent the differences of the declination and

Table 2. Cretaceous paleomagnetic poles in the Arica region.

Rock unit	Locality		Pole					Rot.	Flat.
	Lat. (°S)	Long. (°W)	Lat. (°S)	Long. (°E)	dp (°)	dm (°)	A_{95} (°)	(°)	(°)
Atajaña Formation (AR24)	18.6	70.3	74.7	37.9	2.4	4.4		-13.6 ± 9.5	7.9 ± 13.0
Arica dike swarm (AR31–49)	18.6	70.3	77.2	352.4			3.3	-11.1 ± 9.0	-8.5 ± 13.2

Poles are converted to southern hemisphere poles. dp: radius of confidence oval measured in the direction from site toward pole, dm: radius of 95% confidence oval measured perpendicular to dp, A_{95}: radius of 95% confidence circle. Rot.: rotation and the 95% confidence interval, Flat.: flattening and the 95% confidence interval.

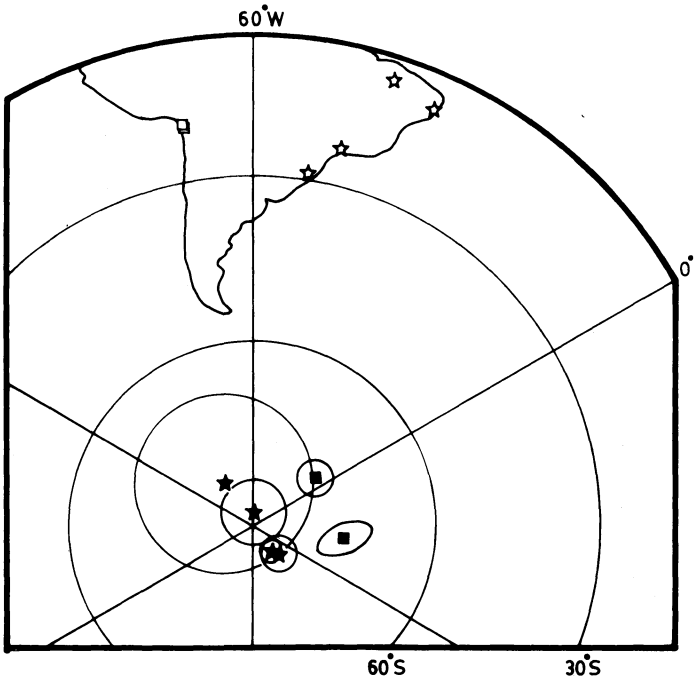


Fig. 6. The Cretaceous paleomagnetic poles of the Arica region (squares) and the stable platform (stars) and their 95% confidence circles/ovals. Corresponding sampling sites are illustrated as small open symbols.

inclination, respectively, between the observed paleomagnetic direction and the expected paleomagnetic direction calculated from the position of the reference pole. The rotation corresponds to the tectonic rotation (a positive rotation value represents the clockwise rotation) and the flattening corresponds to the movement accompanied with the change of the latitude. We calculated these values and their 95% confidence intervals (Table 2) by taking the mean position of the four paleomagnetic poles of the stable platform (88.7°S, 42.0°E, $A_{95}=8.0°$) as the reference. Both of the two Arica region paleomagnetic poles showed significant counterclockwise rotation while no significant values were obtained for the flattening. The results imply that the Arica region underwent about 10°–15° counterclockwise rotation with respect to the platform at least after the Cretaceous period.

The Arica region accurately corresponds to the hinge of the Bolivian orocline, where the northwest-southeast striking Peruvian Andes gradually deflects into the almost north-south striking Chilean/Argentine Andes. At the latitude of the Arica region, the structural trend of the Andes is around N20°W. The rotation values obtained in this study are consistent with the hypothesis proposed by CAREY (1955) and substantiated by us (HEKI, 1983; HEKI *et al.*, 1983; KONO *et al.*, 1984) that the originally linear and north-south trending Central Andes were oroclinally bent forming the present elbow shape at the Peru-Chile border.

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